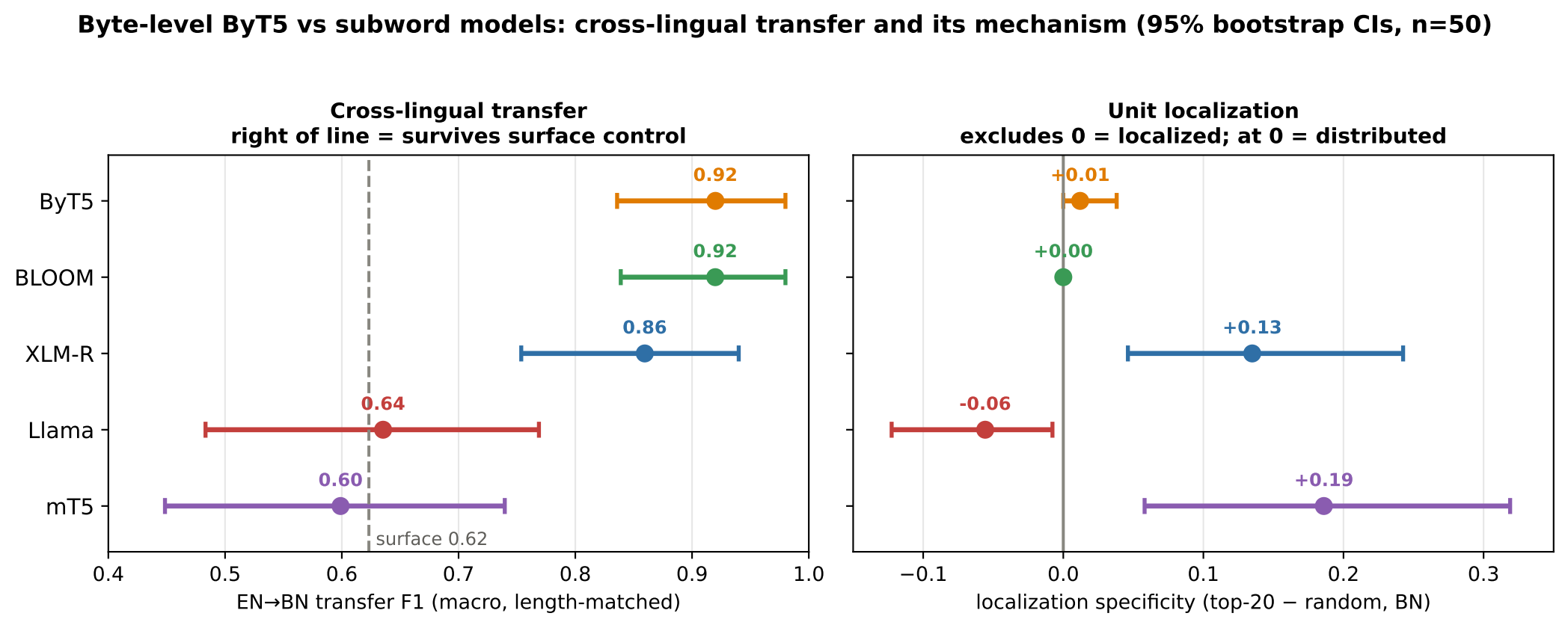
Byte-level ByT5 vs subword models: cross-lingual transfer and its mechanism (95% bootstrap CIs, n=50)
Cross-lingual transfer
right of line = survives surface control
ByT5
BLOOM
XLM-R
Llama
mT5
0.92
0.92
0.86
0.64
0.60
surface 0.62
0.4
0.5
0.6
0.7
0.8
0.9
1.0
EN→BN transfer F1 (macro, length-matched)
Unit localization
excludes 0 = localized; at 0 = distributed
+0.01
+0.00
+0.13
-0.06
+0.19
−0.1
0.0
0.1
0.2
0.3
localization specificity (top-20 − random, BN)